Presentation Skills Evaluation with AI Proposal

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Proposal Version	Date	Reason for Change
1.0	12-October-2024	Proposal First version's specifications are defined

Table 1: Document version history

GitHub: `https://github.com/ArwaaMamdoh23/Presentation-Skills.git`

Abstract

AI is now considered a vital tool in today's world, assisting individuals in solving various problems across different domain. Help them to improve their skills in many things, As we know that creating an effective presentation is crucial for everyone, as it is essential for anyone who wants to leave a good impression and motivate their viewers. Having strong presentation skills is essential for both personal and professional success because everyone desires to communicate their ideas clearly and effectively. We have decided to leverage AI technology to help individuals deliver effective presentations. All users have to do is to film their presentation and then submit the video to our app. Following that, the AI evaluates how they did and gives expert advice, along with a rating that assesses their abilities in presenting. This new method is designed to give users the ability to enhance their presentation skills and leave a lasting impression on their audience.

1 Introduction

1.1 Background

Universities and schools always teach and encourage students to create professional presentations which will help them in their future professions. Therefore, we chose to develop an app that is important for both students to improve their presentation skills and also for workers to refine their techniques, This app emphasizes that successful presentations require more than just well-organized content and a good speech. Factors like nonverbal cues , such as body language, facial expressions, and vocal tone, are crucial in capturing the audience's interest and displaying confidence. However, identifying and improving these elements can be challenging without guidance. So We created a AI-driven app that offers comprehensive feedback on presentation skills. Users, whether they are students or professionals, have the ability to upload a video of their presentation for the AI to analyze, assessing aspects such as body language, facial expressions, voice tone, and speech delivery. The system offers a professional evaluation for every aspect, aiding users in improving their skills and becoming better presenters. This tool is crucial for individuals wanting to enhance their ability to communicate, whether they are getting ready for academic presentations, suggesting ideas at work, or conducting significant meetings.

1.2 Motivation

1.2.1 Academic

Presentations are a powerful tool for communicating ideas, sharing discoveries, and contributing to meaningful discussions in conferences, seminars, and classrooms. They are also a way to leave a lasting impact on the audience. However, many people find it difficult to master these skills, particularly the nonverbal ones like body language and facial expressions, which are crucial for engaging an audience, establishing credibility, and delivering a compelling message. In different academic and professional settings, having strong presentation skills is essential for individuals. Despite their significance, nonverbal communication skills are frequently ignored or challenging to develop using conventional approaches. This difference emphasizes the need for creative, easily accessible tools that offer tailored advice and feedback. By providing focused assistance to people looking to enhance their presentation abilities, this AI-powered app aims to close this gap. The app uses modern technology to analyze and improve important features like posture, facial expressions, and speech delivery. It ensures that users can present with confidence and influence in any situation by enabling them to practice effectively, build confidence, and succeed in their studies and careers.

1.2.2 Business

In the business world, effective presentation abilities are important for professionals who want to share their thoughts convincingly, influence stakeholders, and facilitate important decision-making procedures. The ability to effectively communicate can greatly affects results, whether it be presenting a project proposal, leading a training session, or pitching a new product. And despite that, professionals do not have the chance to get professional feedback that could assist them improve their presentation skills, which could affect thier results, and don't propose thier ideas in a clear way, Our app uses AI power to offer customized insights that improve users' presentation skills, confidence, and effectiveness in business interactions. This instrument is created to aid people at every career stage in their ongoing growth, ultimately leading to increased achievement in their professional lives.

1.3 Problem Statement

Lack of objective feedback on nonverbal communication is one of the biggest obstacles to mastering public speaking and presentations. This makes it hard to assess and enhance facial expressions and body language. People frequently lack access to professional presentation coaching, which prevents them from receiving knowledgeable advice to improve their abilities. Many also struggle to choose the right words and express themselves effectively, which makes it harder for them to express themselves clearly. Furthermore, a lot of people are unaware of how important vocal tone variability is for keeping the audience interested and giving an engaging speech.

2 Project Description

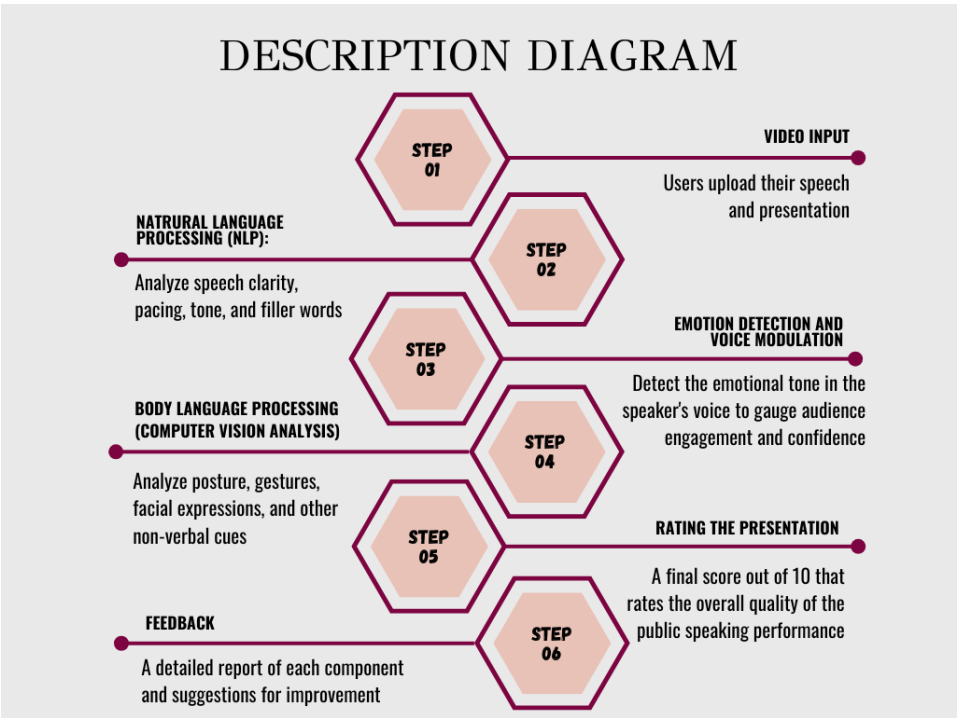

Figure 1: Description Diagram

The Presentation Skills Evaluation project focuses on developing a mobile app powered by AI to help users enhance their public speaking abilities. The app will assess key presentation elements such as speech clarity, body language, and facial expressions. Additionally, it will evaluate how users interact with their audience and respond to questions during presentations. By offering personalized feedback and tracking progress over time, the app aims to serve as a robust resource for improving communication skills, utilizing advanced AI techniques to deliver insights.

2.1 Objectives

- Enhance Public Speaking Skills: Improve overall speaking abilities, focusing on delivery and body language.

- Comprehensive Analysis: Evaluate key presentation elements like speech clarity, facial expressions, and audience interaction.

- Personalized Feedback: Delivering tailored insights to help users recognize and improve specific areas of their performance.

- Progress Tracking: Enable users to monitor their development through progress reports.

- User-Friendly Experience: Develop an intuitive mobile app that simplifies recording and reviewing presentations.

- Interactive Learning: Facilitate practice with real-time audience interactions to simulate real-world scenarios.

2.2 Scope

The System will Contain:

- Uploading Functionality: Users can upload presentations via the mobile app.

- Speech Analysis: Analyzes speech clarity, pacing, tone, and filler words using NLP.

- Body Language Assessment: Evaluates gestures and facial expressions through computer vision.

- Audience Interaction Evaluation: Assesses responses to audience questions.

- Personalized Feedback: Generates tailored feedback reports.

- Progress Tracking: Allows users to monitor improvement with visual reports.

2.3 Project Overview

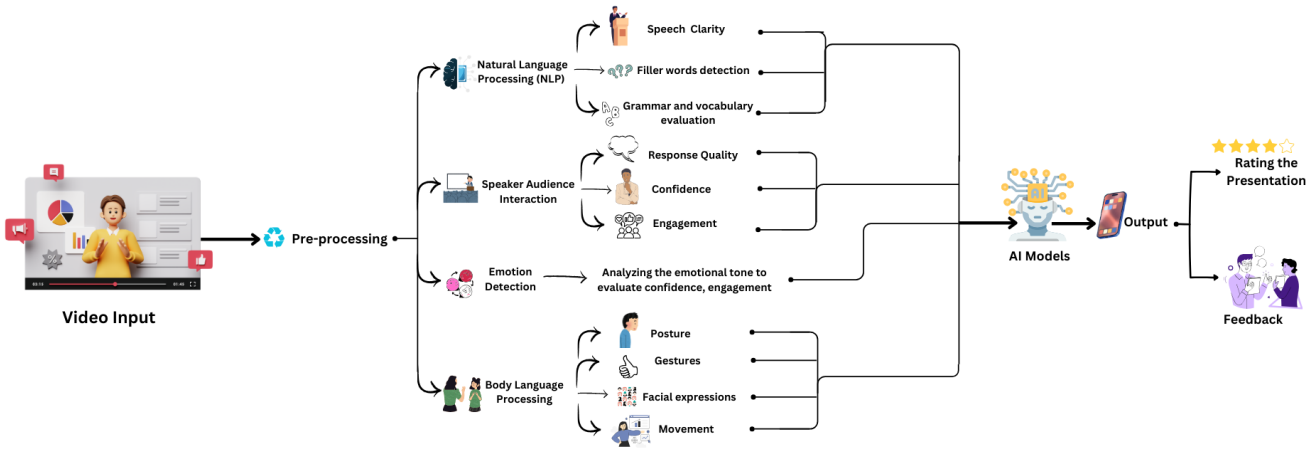

Figure 2: Overview Diagram

- **Collecting the dataset**:

  The dataset for the Presentation Skills Evaluation Project will include various types of data.Such as audio recordings of speeches, video recordings for analyzing body language, and possibly motion capture data to assess gestures and movements.This dataset will help the AI model assess factors like vocal clarity, speech pacing, tone, body posture, eye contact, and hand gestures, providing a comprehensive evaluation of public speaking skills. The dataset has to be split into 70% training and 30% testing. The training set is used to develop the machine learning and NLP models, while the testing set evaluates system performance.

- **Applying NLP and Deep Learning techniques**:

  – **NLP techniques:** We intend to apply NLP to analyze the user's speech: The system evaluates several aspects of speech during presentations.

    • Speech clarity focuses on articulation, pronunciation, and enunciation to determine how clearly the speaker communicates and Pacing examines the speed of delivery to ensure it is appropriate for audience comprehension.

    • Tone analyzes emotional nuances, pitch, and volume, which reveal the speaker's attitude. Finally, the system identifies filler words like "um" or "ah," which can detract from the message's effectiveness. These evaluations contribute to a comprehensive assessment of presentation skills.

  – **Deep learning techniques:** We intend to apply Convolutional Neural Networks (CNNs): Used for analyzing visual features in video frames, such as facial expressions and gestures, to assess body language and Recurrent Neural Networks (RNNs): Useful for processing sequential data, particularly for analyzing speech patterns and pacing over time.

- **Computer Vision** :

- Body Language Processing uses Computer Vision to evaluate a speaker's non-verbal communication. It assesses posture, gestures, facial expressions, and other non-verbal cues like eye contact and proximity to the audience.
- Emotion Detection and Voice Modulation analyzes the emotional tone and vocal characteristics of the speaker. This includes identifying emotions conveyed through voice and evaluating pitch, volume, and tone to gauge audience engagement and confidence. These analyses together enhance the understanding of a speaker's effectiveness and engagement.

- **Performance evaluation**: The performance of each approach is evaluated by calculating accuracy, precision, recall, and F-measure. The best approach will be implemented as a mobile application.

2.4 Stakeholder

2.4.1 Internal

The internal stakeholders consist of the team members, each assigned a specific responsibility. The team members and their respective roles are as follows:

Team Member Name	Responsibilty
Arwaa Mamdoh	Project Manager
Mayar Adel	Flutter
Mostafa Wael	Preprocessing
Ahmed Yehia	Test and train of the dataset

2.4.2 External

- **Students:** Focused on improving public speaking skills for academic success.
- **Professionals:** Aiming to strengthen their presentation skills to excel in the workplace and advance their careers.
- **Public Speakers:** Focused on improving delivery, connecting with audiences, and enhancing overall stage presence.
- **Educational Institutions:** Universities and schools looking to incorporate this tool into their public speaking courses.
- **Organizations:** Businesses aiming to train employees in effective communication and presentation skills.

3 Similar System

3.1 Academic

- **Flink & Larsen** mentioned in this paper [6] that he was using PitchVantage, an AI program that offers real-time feedback on students' vocal delivery skills during presentations by comparing their

recordings to a database of over 10,000 expert-evaluated samples. Improvements in key skills were significant, including pitch variability (from 12.35 to 18.38) and pauses (from 10.06 to 25.98), with a p-value of 0.0001. These findings indicate that AI effectively enhances presentation abilities and self-efficacy,

- **Almubarak, Alhalabi, Albidewi & Alharbi** mentioned in the paper [17] the creation of teacher evaluation system based on T-KPIs and Vision 2030, using two algorithms: one for overall teacher performance and another for specific elements. AI techniques provide data-driven analysis, with evaluative weights determined through expert workshops. Performance elements are assigned percentage weights, such as 100% for "Job Performance," and 6% for "teacher commitment to standard language." Expert input ensures a balanced, adaptable evaluation process that fits various educational settings.

- **Ocho & Dominguez** mentioned in the paper [7] explore the RAP (Real-time Adaptive Feedback) system, which uses various algorithms for multimodal learning analytics (MmLA) but does not specify a particular one. It achieved an accuracy of 85%in evaluating oral presentation skills, with 81% of students recalling positive feedback, increasing to 84% in follow-ups. However, self-assessment of improvement decreased from 74% to 68%. The RAP system likely collects data from video, audio, and sensors. Although it may incorporate algorithms such as Support Vector Machines (SVM) or Recurrent Neural Networks (RNN).

- **Rusmiyanto, Huriati, Fitriani & Tyas** mentioned in [15] that the the project targets a literature review that explores how Artificial Intelligence (AI) enhances English language learners' communication skills across speaking, listening, reading, and writing. It highlights the effectiveness of AI tools, such as speech recognition systems, virtual tutors, adaptive learning platforms, and chatbots, which utilize machine learning and natural language processing (NLP) techniques to improve skills through personalized learning experiences.

- **Zou, Du, Wang, Chen & Zhang** in this paper [16] looked into how well AI speech evaluation tools can help EFL students become more proficient speakers. Significant improvements were noted by the participants; 80% noted gains in pronunciation, 75% in fluency, and 87.5% in reading aloud. 32.5% of users engaged with the programs daily, and 37.5% did so several times a week, indicating frequent usage. Popular applications like Liulishuo, IELTS Liulishuo, EAP Talk, and Shanbay were among the AI models utilized. Concerns regarding the accuracy of the feedback, however, were voiced, suggesting a discrepancy between the outcomes and expectations. To improve user experience, the study suggested tracking progress and increasing the accuracy of feedback. Although the programs were successful overall, resolving feedback issues is essential to maximizing their advantages.

- **Klaassen, Vries & Ceulemans** in [5] compare feedback from a virtual reality (VR) application to traditional feedback from lecturers and peers in order to determine how effective the VR application is for helping students improve their presentation skills. It was discovered that 54% of students thought the VR feedback was "helpful," 62% thought it was "useful," and 50% thought it was "supportive." With an average score of 3.21, the VR group also demonstrated a propensity to contest feedback from the VR system. Wray's RISE model was employed to gauge reflection, and a validated questionnaire

was used to gauge acceptance of feedback. In general, it was found that traditional feedback methods were more effective than VR feedback, which emphasizes the need for more research to maximize VR's educational potential.

- **Taheri & Tan** talk in [19] about creating a safe, virtual reality-based dynamic pitch simulation that incorporates generative AI to improve presenting abilities. Its main objective is to give users realistic feedback so they can practice and improve their skills, especially in high-pressure scenarios like interviews. For tailored feedback, the system makes use of generative AI, open-source language models like LLaMA-2, and speech-to-text conversion. It provides adjustable settings and practical practice opportunities, but no precise accuracy metrics are offered. The study shows how combining VR and AI technologies can enhance presentation training; however, more investigation is required to obtain empirical performance data.

- **Bahreini, Nadolski & Westera** explore in [3] how the FILTWAM framework—which integrates facial and vocal emotion recognition—can improve communication skills training via the use of a serious game known as the Communication Advisor. Real-time feedback on simulated emotions is provided by the system, which leads to notable gains in performance. Vocal emotion recognition was accurate at 61.5%, while facial emotion recognition was 70%. All things considered, the results show that emotion recognition technologies can improve communication skills training by offering trustworthy feedback.

- **Younis, Mohsen, Houssein & Ibrahim** in [20] provide an extensive overview of various machine learning (ML) and deep learning (DL) algorithms employed in the field of emotion recognition, along with their reported accuracies. Notable algorithms include Naive Bayes (NB), which achieved an F1 score of over 90% in binary classification and an average accuracy of 74%; Support Vector Machine (SVM), with an average accuracy of 85.47% and a maximum of 78.97% in another study; and Random Forest (RF), which reported an impressive accuracy of 95.6%. Convolutional Neural Networks (CNN) reached an accuracy of 80%, while in a hybrid model, it achieved an F1 score of 80.76%. Decision Tree (DT) had an average accuracy of 74%. Among deep learning techniques, the Bidirectional Gated Recurrent Unit (Bi-GRU) achieved an accuracy of 79.46%, and a hybrid model combining CNN, Bi-GRU, and SVM yielded an F1 score of 81.27% and an accuracy of 80.11%. Additionally, ensemble learning methods demonstrated significant potential, with the best accuracy reaching 98.2% through stacking techniques. The review highlights the advancements in emotion recognition technologies and their promising applications across various fields.

- **Tarnowski, Kołodziej, Majkowski & Rak** in [4] present a study on recognizing seven emotional states—neutral, joy, sadness, surprise, anger, fear, and disgust—based on facial expressions. The authors employed two classification algorithms: the k-Nearest Neighbors (k-NN) and Multi-Layer Perceptron (MLP) neural networks. The study achieved impressive classification accuracy, with the k-NN classifier reaching 96% for random data division and 63% for subject-independent classification, while the MLP classifier achieved 90% and 73% accuracy, respectively, under similar conditions. These results highlight the effectiveness of using facial expression analysis combined with advanced algorithms for emotion recognition.

- **Mansoorizadeh & Charkai** present in [1] a novel approach for multimodal emotion recognition that integrates facial expressions and speech prosody through an asynchronous feature level fusion method. This approach effectively addresses the natural asynchrony and varying scales of different modalities, leading to significant improvements in emotion recognition rates. The study reports that the unimodal face-based system achieved recognition rates of approximately 44% and 37% on the TMU-EMODB and eNterface'05 databases, respectively, while the speech-based system reached about 60% and 33%. The proposed asynchronous feature level fusion method demonstrated enhancements of about 5% and 12% over decision-level fusion for the same databases, showcasing its superiority in capturing the complementary information from both modalities. Overall, the results indicate that multimodal integration significantly outperforms unimodal systems in emotion recognition task.

- **Guo** in paper [2] investigates the effectiveness of video recordings in improving graduate students' presentation skills through a mixed-methods approach, incorporating peer assessments, instructor evaluations, and external reviewer assessments. While the study primarily focuses on qualitative insights rather than specific algorithms, it emphasizes the use of a rubric adapted from previous research to evaluate presentation components such as organization, supporting evidence, and conclusion. The findings indicate that students' self-assessment scores were significantly lower than peer assessment scores, suggesting a stricter self-evaluation process. The study concludes that video recordings are a valuable tool for enhancing verbal and non-verbal communication skills, organization, and audience engagement, although it does not provide specific accuracy metrics for any algorithms used in the analysis

- **Priolo & Lo Re** in [14] evaluated PitchVantage's AI tool for its impact on improving students' oral presentation skills. The group using PitchVantage scored 4.73, while the control group scored 4.14, reflecting a 14.2% improvement. Students favored combining AI practice with live presentations, noting increased confidence. The results indicate that PitchVantage effectively enhances presentation skills.

- **Dey & Jana** in [12] used Interval Type-2 Fuzzy Logic (IT2FL) to assess the convincing ability of pre-service management students by integrating verbal and non-verbal communication. IT2FL processes input from the Presentation Evaluation Scale (PES) and applies fuzzy logic rules for evaluation.The results show that IT2FL outperforms traditional methods, with an RMSE of 0.05187 and an R^2 of 0.9815, offering a more precise and reliable assessment of presentation skills.

- **Chen, Lai, Chan, Man & Chan** revealed in [10] that they used AI-Assisted presentation training platform which provides us with real time feedback on our students presentation skills. Various aspects of speech were analyses (fluency, pronunciation accuracy, facial expressions, and vocal fillers) comparing the presentations of students to a database of expert-rated samples. We have seen improvements in Fluency with accuracy scores showing a correlation coefficient of 0.136 and pronunciation with 0.450. Finally, these results showed medium to low correlation between humans and Ai showing that AI can help us efficiently. However, they still face some limitations in matching human-level evaluations in fluency assessment and emotion recognition

- **Cherner, Alex, Hou & Haplin** revealed in [11] that they used PitchVantage, an AI powered presentation platform that provides us with real time feedback on oral presentation skills and eye-contact.

Improvements can be seen in Pitch variability though no significant changes were observed over time p=0.89 and eye-contact where students found a loophole where you can just look at the ceiling which results in higher scores. Finally, the results showed that the Ai tool that was used didn't provide significant improvements showing us that it suffers while giving accurate feedback on complex aspects like emotion recognition.

- **Sakkali, Ginkel & Meijboom** revealed in [8] that they used Honest Mirror AI which provides real time feedback on presentation skills. Their system analyses various aspects like delivery,posture,gestures and body language by comparing the student's recordings to a database of expert rated samples. Decent improvements were seen on posture and gestures through actionable feedback. the results have showed medium level correlation between AI feedback and an expert which shows that Honest Mirror have succeeded in providing effective feedback even though it may have some limitations

- **Dimitriadou & Lanitis** revealed in [18] that they used BLA which is an AI tool that provides real time feedback on educator's body language during lectures. This mode analyses various aspects of body language like facial expression, and movements and speech metrics through the comparison of the lecturer's recording to indicators based on literature and expert interviews. Improvements can be seen in facial expressions with an improvement from a mean score of 0.165 in the first lecture to 0.341 in the second lecture. while hand movements show an increase from 0.125 to 0.396. The BLA model showed significant improvements while assessing non-verbal cues. However it showed some limitations when it comes to addressing speech quality.

- **Mulyani, Filzafatin Habibah & Mukarom** in [13] highlight the value of presentation skills in learning Arabic, highlighting how they contribute to international communication and fitting in with Indonesia's Outcome Based Education (OBE) model. It draws attention to how incorporating AI technologies, such as Xembly and SemblyAi, can increase these abilities, as seen by studies showing that AI can help students become more proficient communicators. The sorts of AI that can be used, implementation tactics, and related opportunities and problems in Arabic language teaching are all examined in this study using qualitative data analysis.

- **Shin, Choi, Hwang & Kim** in [9] examined how voice-based intelligent system (VIS) design parameters, such as the number of trials required to get the correct answer and the response's sentence structure, affect the emotional satisfaction of the user. Based on these variables, researchers evaluated emotional satisfaction levels using Analysis of Variance (ANOVA) and Tukey analysis.The results showed that the number of trials and sentence structure had a significant impact on emotional satisfaction, which was categorized into four levels for reliability and three for pleasurability. One trial and a distinct reference resulted in the highest level of satisfaction. The study found strong statistical significance with p-values of less than 0.001 for the number of trials, 0.004 for pleasurability, and 0.007 for reliability, despite the lack of specific accuracy percentages.

3.2 Business Applications

- **Orai** is an AI-driven voice coaching app designed to enhance public speaking skills by providing instant feedback. Users can practice their speeches, and the app analyzes various aspects such as tone, pitch, volume, clarity, and the use of filler words.

Figure 3: Orai

- **Speeko** focuses on helping users refine their public speaking skills through voice recognition technology. The app provides feedback on pacing, intonation, and articulation while offering a range of challenges to practice different speaking scenarios.

Figure 4: Speeko

- **Virtual Speech** combines online learning with virtual reality (VR) training environments to simulate real-world speaking scenarios. Users can practice in various settings with virtual audiences, which helps reduce anxiety and improve delivery.

Figure 5: Virtual Speech

4 What is new in the Proposed Project?

Most available projects in the field today like speechiy,tally and orai are either focused on linguistic abilities (speech,clarity,grammar,vocabulary and pace) or non-verbal cues (body language, facial expressions and eye contact).Our project on the other hand will combine both these crucial aspects into one comprehensive tool. These current systems use simpler AI models that focus on a few aspects of presentation skills mostly linguistic while our project will use much more developed Ai and deep learning models that allows for a much deeper understanding of the various aspects of the presentation at hand. Lastly, most of the current tools provide very generic feedback. Our project will provide detailed feedback on some mistakes and suggestions for improvement as well as a scaled rating for body language. Accurate measurement on pace (words per minute) in different sections of the presentation and speech clarity, tone and various other aspects.

5 Proof of concept

- **Speech Analysis** :Libraries like pydub were used for handling audio files and including timestamps and SpeechRecognition was used for speech to text transcription.

  - **Grammar check**:The T5-Base-Grammar-Correction which is a deep learning model for NLP developed by google research was used to return a version of the text transcription which is grammatically perfect.

  - **Grammar analysis**:The original transcription and the grammatically correct version of the string have been normalized by removing punctuation and capitalization.A comparison has been made between both of them to calculate a ratio of errors to the length of the string.Based on this ratio, an accurate grammatical score will be returned reflecting the grammatical ability in the speech.

  - **Speech pace**: Using the timestamps placed by pydub and the transcription from SpeechRecognition we can easily calculate how many words are spoken each minute which will allow us to give appropriate feedback.

```python
# Load the audio file using pydub to get duration
audio_segment = AudioSegment.from_file(audio_file_path)

# Get the duration of the audio file in seconds
audio_duration = len(audio_segment) / 1000  # pydub returns duration in milliseconds

# Load the audio file for the speech recognizer
audio_file = sr.AudioFile(audio_file_path)

# Process the audio file in chunks
with audio_file as source:
    recognizer.adjust_for_ambient_noise(source, duration=1)  # Optional: helps with noisy audio

    # Define chunk size in seconds (e.g., 30 seconds per chunk)
    chunk_size = 30

    transcription = ""  # To store the full transcription
    current_position = 0

    # Loop through the audio in chunks
    while current_position < audio_duration:
        try:
            # Record the next chunk
            audio_chunk = recognizer.record(source, duration=chunk_size)

            # Transcribe the chunk
            chunk_text = recognizer.recognize_google(audio_chunk)
            transcription += chunk_text + " "  # Append chunk transcription
            print(f"Transcribed chunk {current_position}-{current_position+chunk_size}s: {chunk_text}")
        except sr.UnknownValueError:
            print(f"Google Speech Recognition could not understand the audio between {current_position}s and {current_position+chunk_size}s")
        except sr.RequestError as e:
            print(f"Could not request results from Google Speech Recognition service; {e}")

        # Update position to move to the next chunk
        current_position += chunk_size
```

Figure 6: Demo 1

```python
# Load the pre-trained T5 model for grammar correction
model = T5ForConditionalGeneration.from_pretrained("vennify/t5-base-grammar-correction")
tokenizer = T5Tokenizer.from_pretrained("vennify/t5-base-grammar-correction")

model = T5ForConditionalGeneration.from_pretrained("vennify/t5-base-grammar-correction")
tokenizer = T5Tokenizer.from_pretrained("vennify/t5-base-grammar-correction")

def correct_grammar(sentence, max_length=512):
    corrected_text = ""
    i = 0
    while i < len(sentence):
        # Get the next chunk of the sentence
        chunk = sentence[i:i + max_length]
        # Prepare input with prefix "grammar: " for T5 model
        input_text = "grammar: " + chunk
        inputs = tokenizer.encode(input_text, return_tensors="pt", max_length=max_length, truncation=True)

        # Generate corrected text
        outputs = model.generate(inputs, max_length=max_length, num_beams=4, early_stopping=True)
        corrected_chunk = tokenizer.decode(outputs[0], skip_special_tokens=True)

        # Append the corrected chunk
        corrected_text += corrected_chunk + " "

        # Move to the next chunk
        i += max_length

    return corrected_text.strip()
```

Figure 7: Demo 2

```python
# Function to normalize text by removing punctuation and converting to lowercase
def normalize_text(text):
    # Remove punctuation and convert to lowercase
    text = text.translate(str.maketrans('', '', string.punctuation)).lower()
    return text
def grammatical_score(original, corrected):
    # Normalize both the original and corrected sentences
    original_normalized = normalize_text(original)
    corrected_normalized = normalize_text(corrected)

    # Using difflib to compare the normalized sentences
    diff = difflib.ndiff(original_normalized.split(), corrected_normalized.split())
    changes = list(diff)

    # Count the number of words that changed (added/removed/modified)
    corrected_words = [word for word in changes if word.startswith('+ ')]
    removed_words = [word for word in changes if word.startswith('- ')]

    num_changes = len(corrected_words) + len(removed_words)
    total_words = len(original_normalized.split())

    # Calculate the error ratio (number of corrections divided by total words in the original text)
    error_ratio = num_changes / total_words if total_words > 0 else 0

    # Score based on the error ratio, normalizing it with respect to the length
    if error_ratio == 0:
        return 10  # Perfect grammar
    elif error_ratio < 0.05:
        return 9  # Very few errors (minor grammar issues)
    elif error_ratio < 0.1:
        return 8  # A few grammar issues
    elif error_ratio < 0.2:
        return 6  # Noticeable grammar issues
    else:
        return 4  # Many grammatical errors
```

Figure 8: Demo 3

```python
def calculate_speech_pace(audio_file):
    # Load the audio file using pydub to get the duration
    audio_segment = AudioSegment.from_file(audio_file)
    # Get the duration of the audio in seconds
    duration = len(audio_segment) / 1000  # pydub returns duration in milliseconds

    # Transcribe audio to text using SpeechRecognition
    recognizer = sr.Recognizer()
    with sr.AudioFile(audio_file) as source:
        audio_data = recognizer.record(source)
        try:
            transcription = recognizer.recognize_google(audio_data)
        except sr.UnknownValueError:
            print("Speech Recognition could not understand audio")
            return None
        except sr.RequestError:
            print("Could not request results from Google Speech Recognition service")
            return None

    # Count the number of words in the transcription
    word_count = len(transcription.split())
    # Calculate pace in words per minute (WPM)
    minutes = duration / 60
    pace = word_count / minutes if minutes > 0 else 0
    # Determine pace category
    if 130 <= pace <= 160:
        pace_feedback = "Your pace is perfect."
    elif 100 <= pace < 130:
        pace_feedback = "You need to speed up a little bit."
    elif pace < 100:
        pace_feedback = "You are going very slow."
    elif 160 < pace <= 190:
        pace_feedback = "You need to slow down a little bit."
    else:  # pace > 190
        pace_feedback = "You are going very fast."

    return pace, transcription, pace_feedback
```

Figure 9: Demo 4

6 Project Management and Deliverables

6.1 Deliverables

- The project will produce a mobile application that give the user a feedback on their presentations focusing on body language, facial expressions, eye contact, speech fluency, grammatical correctness, and overall appearance.Also a various number of papers on topics like AI feedback systems, non-verbal behavioral analysis and NLP based speech analysis will be included,
- **deliverables are**:
  - Proposal document
  - Academic Papers
  - SRS document
  - SDD document
  - Final Thesis document
  - Mobile Application

6.2 Tasks and Time Plan

Task	Start Date	End Date	Duration (days)
Idea and Supervisors	12.7.24	9.9.24	52
Information collection and researches	15.9.24	5.10.24	20
Survey and Proposal Preparation	6.10.24	19.10.24	13
Proposal Evaluation 10%	20.10.24	27.10.24	7
SRS Preparation	1.11.24	21.12.24	50
SDD Preparation	1.1.25	7.8.25	37
SRS/SDD Evaluation 65%	8.2.25	21.2.25	13
Technical Working Implementation	1.3.25	14.4.25	55
Technical Evaluation 80%	14.4.25	21.4.25	7
Test and Validate	22.4.25	7.5.25	15
Final Prototype 90%	7.5.25	15.5.25	7
Final Thesis 100%	21.6.25	29.6.25	8

Figure 10: Time plan table

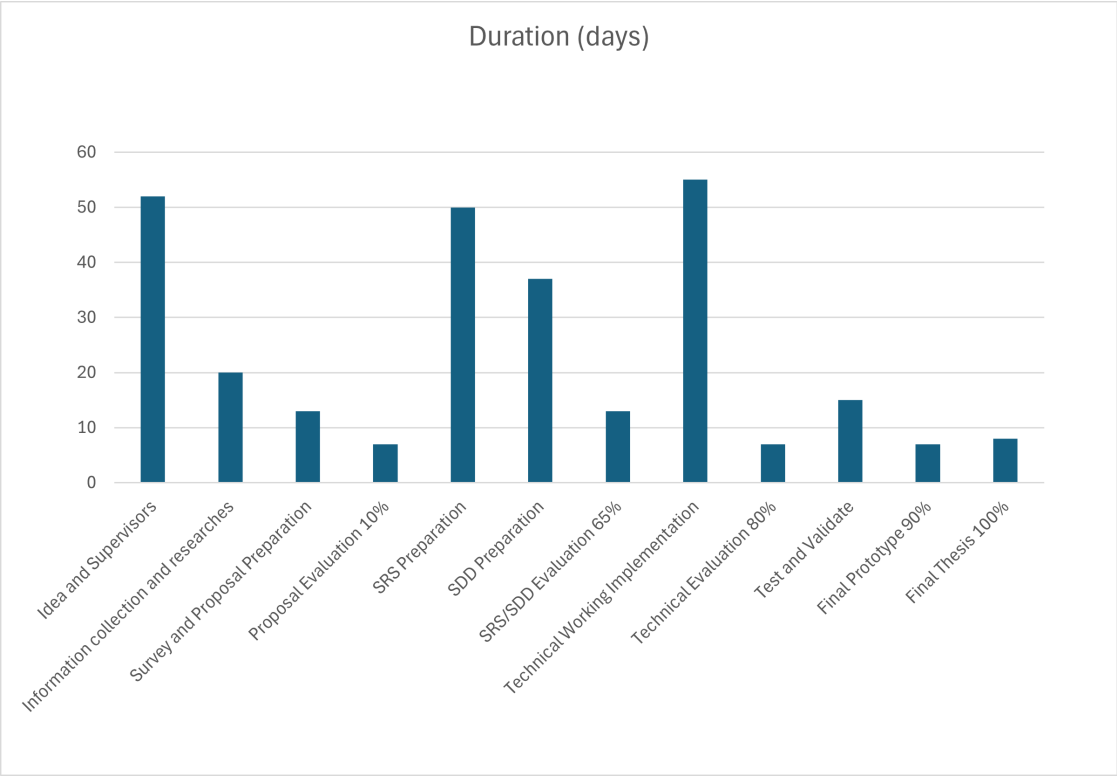

Figure 11: Time plan chart

6.3 Budget and Resource Costs

- A server to run the application

7 Supportive Documents

7.1 Datasets

- **Audio Dataset**: This Dataset is organized into 5 folders, each representing a specific emotion (Angry,Fear,Happy,Sad,Neutral).The dataset is balanced as each folder contains more than 1669 audio files in .wav format. Audios length is from 2 to 5 seconds representing real-life emotional expressions. This dataset is designed for training and evaluating a speech emotion recognition model through voice. The dataset is labelled to accurately detect the intended emotion.

- **Facial Expressions Dataset**: This Dataset is very large yet very organized it contains many facial screenshots from many various people.These pictures are 48 x 48 grey-scale images where the faces are mostly centered in the images . This dataset is split into 2 main files train and test each of these 2 folders contains 7 folders expressing the seven emotions to be predicted(Anger, Disgust, Fear, Happy, Sad, Surprise, Neutral). The training set consists of 28,709 examples and the public test set consists of 3,589 examples.

Figure 12: Dataset Sample

7.2 Survey

This survey examines public speaking habits, difficulties, and preferences based on 101 responses. Although the majority of participants (82.2%) have not used tools for improvement, many indicated interest in features like body language analysis, speech tips, and personalized feedback in an app. The biggest obstacle (42.6%) was nervousness, which was followed by problems with body language and delivery. The majority of respondents, who were between the ages of 18 and 24, stated that professional and academic growth were the main reasons they wanted to improve their abilities. The survey highlights the need for easily accessible resources to boost presentation skills.

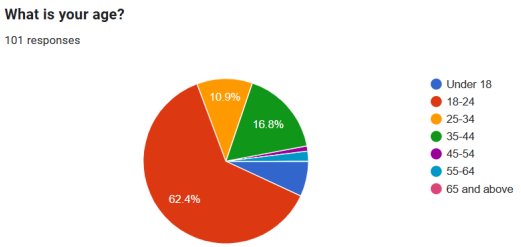

Figure 13: Age

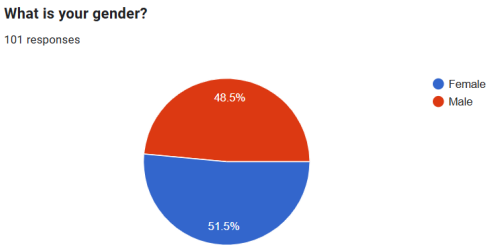

Figure 14: Gender

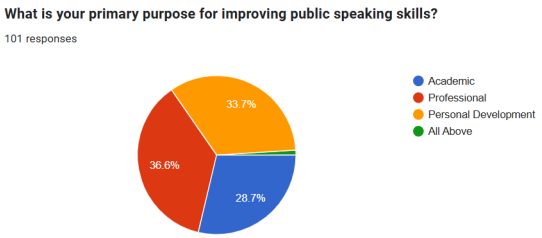

Figure 15: Purpose

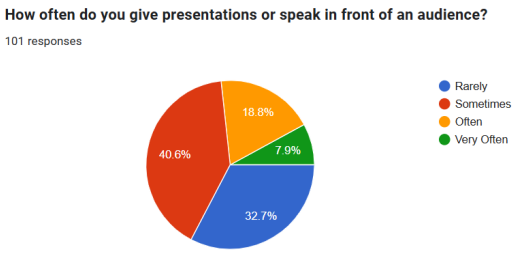

Figure 16: How Often

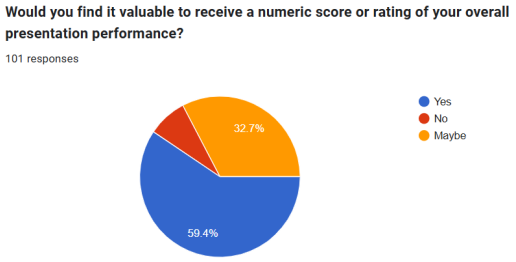

Figure 17: Score

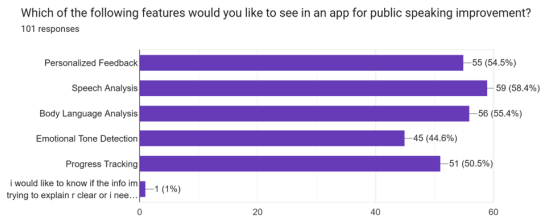

Figure 18: Features

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